

MATH 161A - Applied Probability & Statistics

Connor Petri

Fall 2025

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1. Skipped

2. Probability

2.1 Sample Spaces and Events

Probability

- Probability is used as a mathematical tool to understand or describe random phenomena, chance variation, and uncertainty.
- We will use probability as a tool to evaluate the reliability of our conclusions about the population when we only have sample information.
- Probability is used in a variety of fields including genetics, quality control, and finance.
- Probability is used as a model for situation for which outcomes occur randomly.
- Such situations are called **experiments**.
- Examples:
 - Toss a coin.
 - Roll a die.
 - Draw a card from a deck.

Vocabulary

- An action or process whose outcome is subject to uncertainty is called an **experiment**.
- The possible outcomes of an experiment are called **sample space**.
- An **event** is a specific outcome or a set of outcomes from an experiment.
- A **simple event** is an event that consists of a single outcome from sample space S .

Example What is the outcome of rolling an even (E), rolling greater than 3 (F), and rolling a 2 (G) on a 6-sided die?

$$S = 1, 2, 3, 4, 5, 6$$

$$E = 2, 4, 6$$

$$F = 4, 5, 6$$

$$G = 2$$

A Note on Venn Diagrams All Venn Diagrams in this course require a rectangle to represent the sample space. Events are represented by circles within the rectangle.

Operations

Set Operations

- $A \subseteq B$: A is a subset of B .
- $A \cap B$: the intersection of A and B (elements in both A and B).
- $A \cup B$: the union of A and B (elements in either A or B).
- A^c or A' : the complement of A (elements in S that are not in A).

Logical Operations

- A or B : $A \vee B$
- A xor B : $A \oplus B$
- A and B : $A \wedge B$
- not A : $\neg A$

Mutually Exclusive Events

- Events A and B are **mutually exclusive** if they cannot both occur at the same time.
 - In terms of set operations: $A \cap B = \emptyset$.
 - Example: Rolling a die and getting an even number (E) or an odd number (O) are mutually exclusive events.
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Identities

$$A \cup A' = S$$

$$A \cap A' = \emptyset$$

DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

$$\neg(A \vee B) = \neg A \wedge \neg B$$

$$\neg(A \wedge B) = \neg A \vee \neg B$$